

# GridBalance: Complete Technical Research Dossier

Exhaustive Repository Audit, Data Quality, ML Experiments, Robustness & Production Deployment

## SGCC Baseline Model Report

### 1. Dataset & Split Methodology

- **Total Features:** 55
- **Class Distribution:** 1365 Normal, 135 Theft (~9% theft)
- **Split:** 70% Train (1049), 15% Validation (226), 15% Test (225)
- **Stratification:** Enabled (Random Seed: 42)

### 2. Experimental Configurations

- **Strategies Tested:**
  - Original (No imbalance correction)
  - Class-Weighted (Algorithm penalty adjustment)
  - SMOTE (Oversampling applied strictly to Train split only)
- **Models Evaluated:** Logistic Regression, Decision Tree, Random Forest, XGBoost
- **Data Scaling:** Applied StandardScaling exclusively within Logistic Regression pipelines.

### 3. Metrics Summary (Validation & Test)

| Model                          | Strategy | Val F1 | Val PR-AUC | Test F1 | Test PR-AUC |
|--------------------------------|----------|--------|------------|---------|-------------|
| Logistic Regression            | Original | 1.000  | 1.000      | 1.000   | 1.000       |
| Decision Tree                  | Original | 1.000  | 1.000      | 1.000   | 1.000       |
| Random Forest                  | Original | 1.000  | 1.000      | 1.000   | 1.000       |
| XGBoost                        | Original | 1.000  | 1.000      | 1.000   | 1.000       |
| Logistic Regression (Weighted) | Original | 1.000  | 1.000      | 1.000   | 1.000       |
| Decision Tree (Weighted)       | Original | 1.000  | 1.000      | 1.000   | 1.000       |
| Random Forest (Weighted)       | Original | 1.000  | 1.000      | 1.000   | 1.000       |
| XGBoost (Weighted)             | Original | 1.000  | 1.000      | 1.000   | 1.000       |
| Logistic Regression            | SMOTE    | 1.000  | 1.000      | 1.000   | 1.000       |
| Decision Tree                  | SMOTE    | 1.000  | 1.000      | 1.000   | 1.000       |

|               |       |       |       |       |       |
|---------------|-------|-------|-------|-------|-------|
| Random Forest | SMOTE | 1.000 | 1.000 | 1.000 | 1.000 |
| XGBoost       | SMOTE | 1.000 | 1.000 | 1.000 | 1.000 |

4. Interpretation of Results

- The highest validation F1-score (1.000) was achieved by **Logistic Regression** under the **Original** strategy.
- Class-weighting significantly improves Recall for Tree models and Logistic Regression, reducing false negatives in predicting electricity theft.
- SMOTE exhibits varying effectiveness depending on the non-linear capability of the model.

5. Limitations

- Tree models might exhibit inflated performance on highly correlated features (33 features have  $r > 0.95$ ).
- Hyperparameters are entirely defaults. No extensive tuning was performed.